KAIZEN MATHS

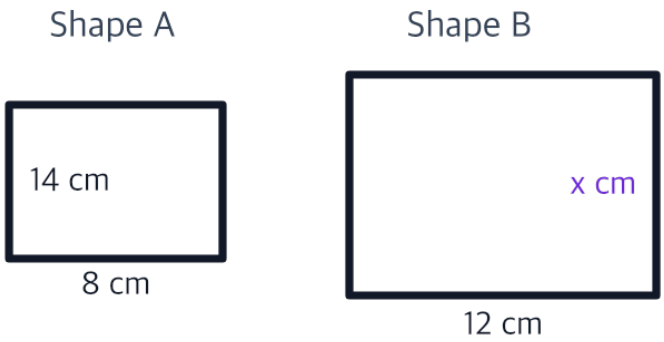
Lesson starter

Scale Drawings and Similar Shapes

Similar Shapes

Name _____

SA1



The two rectangles are similar. Find the missing length x.

GEO-9EE4-L2 · SA

Show your working

KAIZEN MATHS

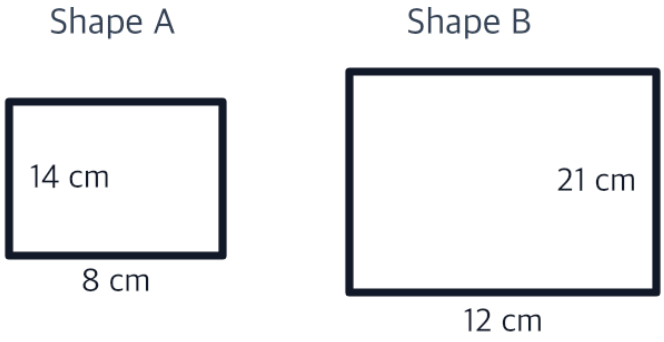
Lesson starter

Scale Drawings and Similar Shapes

Similar Shapes

Name _____

SB1



The two rectangles are similar. Find the scale factor from Shape A to Shape B.

GEO-9EE4-L2 · SB

Show your working

KAIZEN MATHS

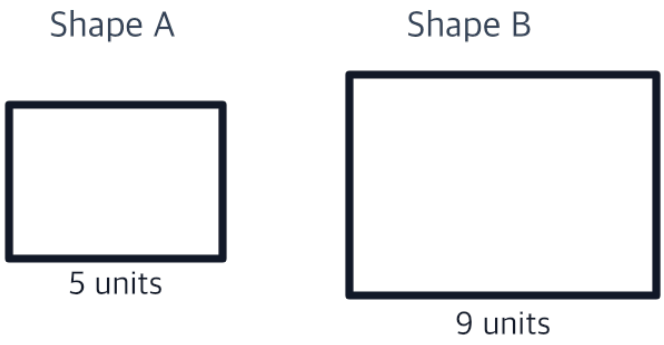
Lesson starter

Scale Drawings and Similar Shapes

Similar Shapes

Name _____

SC1



Two similar shapes have length ratio 5:9. The smaller shape has perimeter 75 cm. Find the perimeter of the larger shape.

GEO-9EE4-L2 · SC

Show your working

KAIZEN MATHS

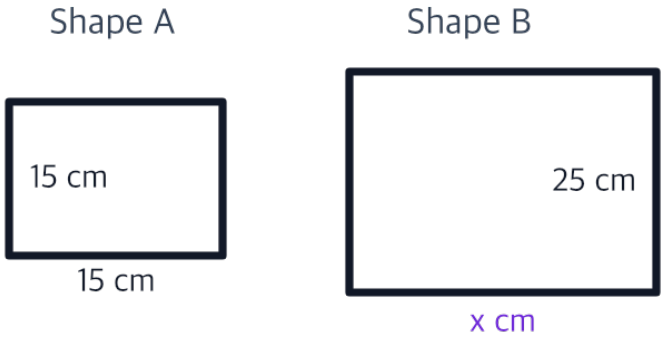
Lesson starter

Scale Drawings and Similar Shapes

Similar Shapes

Name _____

SD1



The two rectangles are similar. Find the missing length x.

GEO-9EE4-L2 · SD

Show your working

KAIZEN MATHS

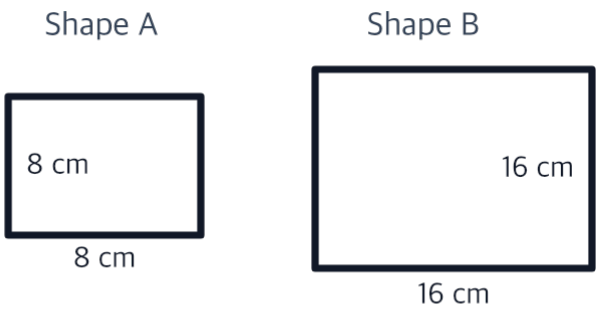
Exit ticket

Scale Drawings and Similar Shapes

Similar Shapes

Name _____

EA1



The two rectangles are similar. Find the scale factor from Shape A to Shape B.

GEO-9EE4-L2 · EA

Show your working

KAIZEN MATHS

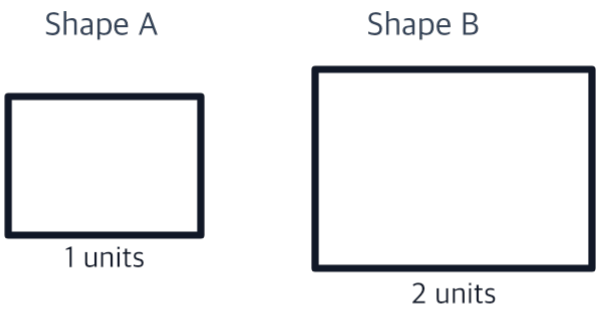
Exit ticket

Scale Drawings and Similar Shapes

Similar Shapes

Name _____

EB1



Two similar shapes have length ratio 1:2. The smaller shape has perimeter 15 cm. Find the perimeter of the larger shape.

GEO-9EE4-L2 · EB

Show your working

KAIZEN MATHS

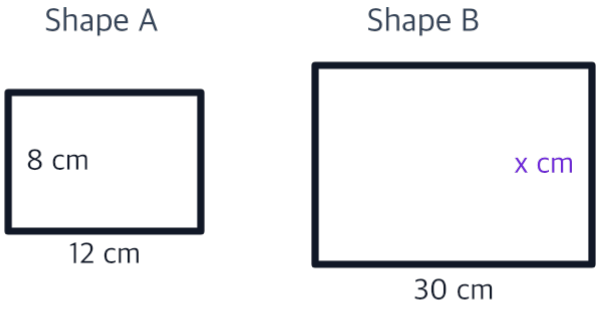
Exit ticket

Scale Drawings and Similar Shapes

Similar Shapes

Name _____

EC1



The two rectangles are similar. Find the missing length x.

GEO-9EE4-L2 · EC

Show your working

KAIZEN MATHS

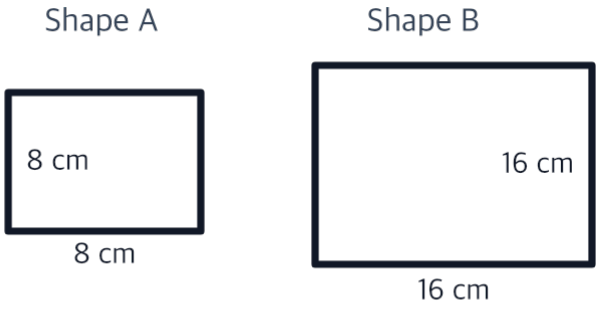
Exit ticket

Scale Drawings and Similar Shapes

Similar Shapes

Name _____

EA1



The two rectangles are similar. Find the scale factor from Shape A to Shape B.

GEO-9EE4-L2 · EA

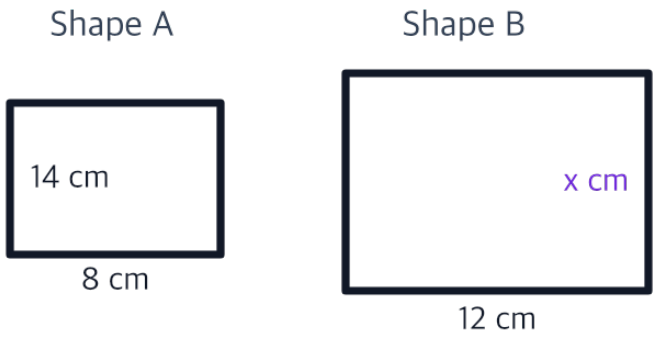
Show your working

Teacher answers & working

Scale Drawings and Similar Shapes · Similar Shapes

GEO-9EE4-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



The two rectangles are similar. Find the missing length x.

Answer:

21 cm

Working:

Find the length scale factor using a pair of corresponding sides.

Scale factor = 12 ÷ 8 = 1.5.

Missing length = corresponding length × scale factor.

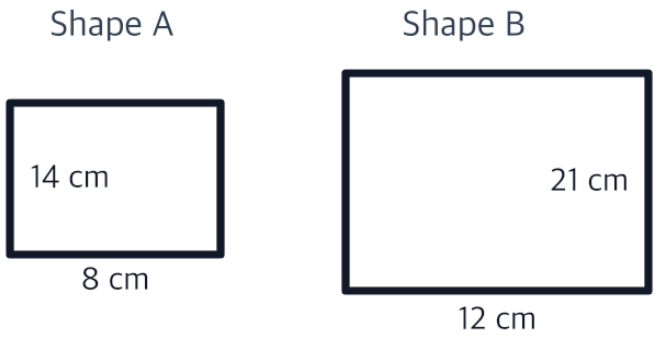
$x = 14 \times 1.5 = 21$ cm.

Teacher answers & working

Scale Drawings and Similar Shapes · Similar Shapes

GEO-9EE4-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



The two rectangles are similar. Find the scale factor from Shape A to Shape B.

Answer:

3:2 or 1.5

Working:

Use any pair of corresponding sides.

Scale factor from A to B = B side ÷ A side.

Scale factor = 12 ÷ 8 = 3/2.

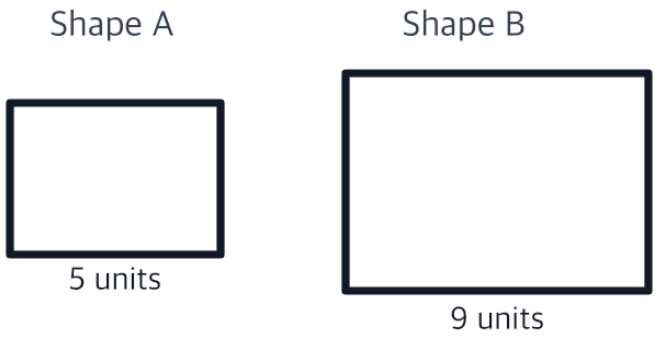
So the scale factor is 3:2, which is 1.5.

Teacher answers & working

Scale Drawings and Similar Shapes · Similar Shapes

GEO-9EE4-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



Two similar shapes have length ratio 5:9. The smaller shape has perimeter 75 cm. Find the perimeter of the larger shape.

Answer:

135 cm

Working:

For similar shapes, perimeter ratio is the same as length ratio.

Length ratio = 5:9, so perimeter ratio = 5:9.

Scale factor from smaller to larger = 9 ÷ 5.

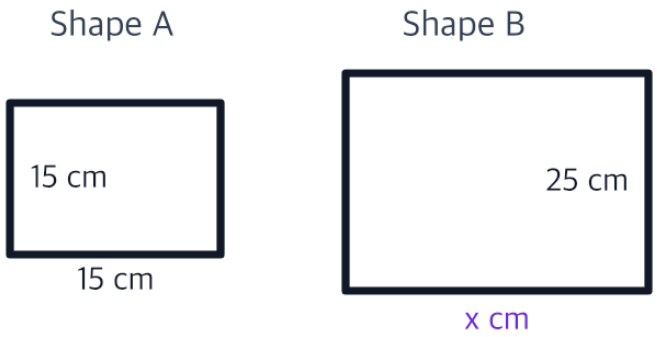
Larger perimeter = 75 × 9 ÷ 5 = 135 cm.

Teacher answers & working

Scale Drawings and Similar Shapes · Similar Shapes

GEO-9EE4-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



The two rectangles are similar. Find the missing length x.

Answer:

25 cm

Working:

Find the length scale factor using a pair of corresponding sides.

Scale factor = 25 ÷ 15 = 1.67.

Missing length = corresponding length × scale factor.

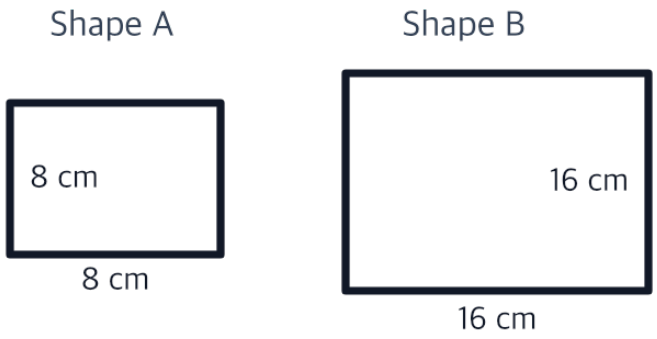
$x = 15 \times 1.67 = 25$ cm.

Teacher answers & working

Scale Drawings and Similar Shapes · Similar Shapes

GEO-9EE4-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



The two rectangles are similar. Find the scale factor from Shape A to Shape B.

Answer:

2:1 or 2

Working:

Use any pair of corresponding sides.

Scale factor from A to B = B side ÷ A side.

Scale factor = 16 ÷ 8 = 4/2.

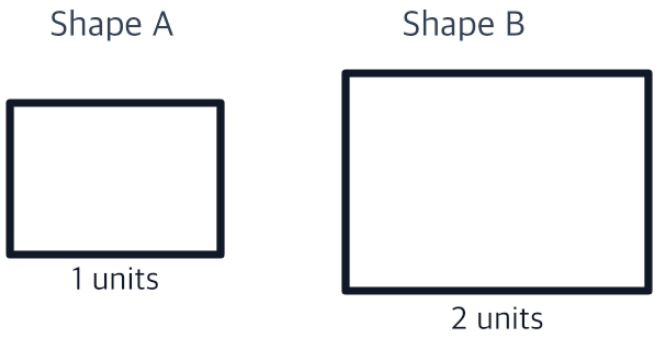
So the scale factor is 2:1, which is 2.

Teacher answers & working

Scale Drawings and Similar Shapes · Similar Shapes

GEO-9EE4-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



Two similar shapes have length ratio 1:2. The smaller shape has perimeter 15 cm. Find the perimeter of the larger shape.

Answer:

30 cm

Working:

For similar shapes, perimeter ratio is the same as length ratio.

Length ratio = 1:2, so perimeter ratio = 1:2.

Scale factor from smaller to larger = 2 ÷ 1.

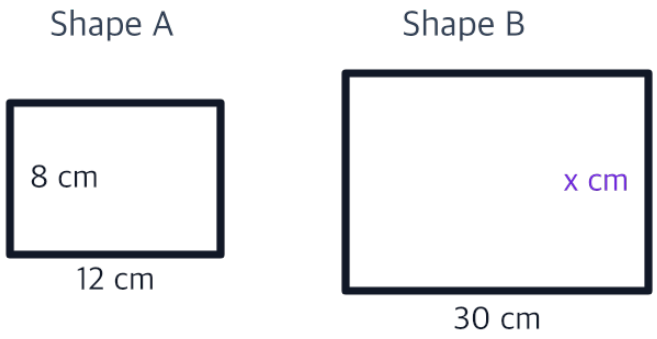
Larger perimeter = 15 × 2 ÷ 1 = 30 cm.

Teacher answers & working

Scale Drawings and Similar Shapes · Similar Shapes

GEO-9EE4-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



The two rectangles are similar. Find the missing length x.

Answer:

20 cm

Working:

Find the length scale factor using a pair of corresponding sides.

Scale factor = 30 ÷ 12 = 2.5.

Missing length = corresponding length × scale factor.

$x = 8 \times 2.5 = 20$ cm.